



NUI MAYNOOTH

Ollscoil na hÉireann Má Nuad

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THE NATIONAL UNIVERSITY OF IRELAND MAYNOOTH

MATHEMATICAL PHYSICS

Year 3

Autumn Repeat Examination

2012 - 2013

Quantum Mechanics I

MP363

Professor P. H. Damgaard, Professor D. M. Heffernan, Professor C. Nash, Dr. P. Watts

Time allowed: $1\frac{1}{2}$ hours

Answer **two** questions

All questions carry equal marks

1. (a) The Rayleigh-Jeans formula for the intensity of radiation produced by a black-body of temperature T is

$$I(\nu, T) = \frac{2\nu^2 kT}{c^2}.$$

Write down Planck's modification of this formula and show that it agrees with the Rayleigh-Jeans formula for small values of the frequency ν .

- (b) Light of wavelength $3 \times 10^{-7}\text{m}$ shines on a sodium sample, which then emits electrons with kinetic energy 1.4eV . When the same sample is illuminated with light of wavelength $4 \times 10^{-7}\text{m}$, it emits electrons with kinetic energy 0.4eV . Use these data to find Planck's constant h and the minimum energy W needed to free an electron from the surface of sodium. (Note: $c = 3 \times 10^8\text{m} \cdot \text{s}^{-1}$.)
2. (a) Let $\mathcal{H}$ be a Hilbert space with inner product $\langle | \rangle$. If A is an operator acting on $\mathcal{H}$ and ψ is a state in $\mathcal{H}$, give the definition of the expectation value $\langle A \rangle_\psi$, and show that if A is Hermitian, then $\langle A \rangle_\psi$ is real.
- (b) Let ψ be a solution to the Schrödinger equation

$$H\psi = i\hbar \frac{\partial \psi}{\partial t}.$$

Show that this implies that

$$\frac{d}{dt} \langle \psi | \psi \rangle = 0$$

and explain why this is necessary in order to impose the normalisation condition $\langle \psi | \psi \rangle = 1$.

3. Let $\{\psi_0, \psi_1, \psi_2, \dots\}$ be an orthonormal set of energy eigenstates with energies $E_n = -\mathcal{E}/2^n$, where $\mathcal{E}$ is a positive constant with units of energy.
- (a) Suppose the system is in the state

$$\Psi = \sum_{n=0}^{\infty} \frac{A}{3^n} \psi_n$$

where A is a positive real number. Find the value of A which ensures that Ψ is normalised to unity.

- (b) Compute the probability that a measurement of the system's energy returns the value $-\mathcal{E}/2$.
- (c) Show that the expectation value of the energy is $-16\mathcal{E}/17$.